

A Dynamic Intelligent Approach Based on Gaussian Function for Prediction of the Flashover Voltage Conditions on Polluted Polymer Insulators

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Abstract—This research presents a dynamic modeling approach to predict flashover voltage (FOV) of polymer insulators. Pre-flashover conditions can be used to represent analytical formulations for different stages of dry band arcing activities on the surface of polluted insulator. In this study, experimental results of pollution flashover tests are used to develop a circuit model of pollution arcs based on Obenaus' model. Estimation of circuit values is done by use of intelligent algorithm with no dependency on profile of insulators which is also more beneficial than typical analytical calculations. Then, separation and investigation of discharge and pollution voltage is done and proper classification is made. This classification is later used to predict pre-flashover conditions and pollution flashover. Gaussian model is used to make fitness of voltage of pollution layer and voltage of discharge on the surface of insulator. Gaussian representation of each stage in pollution flashover contains critical signals leading to complete discharge. Based on the obtained model, a flowchart is presented to predict dynamic stages of flashover. Comparison to various experimental data shows close correlations with the results of proposed model. An online monitoring system can be developed according to obtained results in order to prevent pollution flashover.

Index Terms—Polymer insulator, pollution flashover, dynamic modeling, flashover voltage, leakage current, flashover prediction.

I. INTRODUCTION

ELECTRICAL performance of outdoor insulators is significantly influenced by environmental conditions [1]–[3]. For example, in polluted areas, industrial regions and coastal zones, a conductive layer caused by contaminants in combination of wet conditions can be formed on the surface of insulators [4]. When pollution is combined with moisture, a resistive leakage current (LC) is flowed on the surface of insulator which leads to formation of dry bands [5]. This is accompanied by an increase in partial discharge activities. As LC increases, a partial arc starts to propagate along pollution layer which finally results in complete discharge or flashover of the insulator [6], [7].

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In recent years significant attention has been paid to mechanisms of development of electric arc concerning polluted insulators [8]–[10]. Flashover modeling is usually based on formerly basic Obenaus' model [11], in which flashover is considered as an arc series with a residual resistance including part of pollution layer unabridged by the arc. Pollution flashover models are developed as either static models [12] or dynamic models [13], [14]. Although static models are capable of specifying the critical pre-flashover voltage, but development of arc parameters such as LC or variation of arc impedance are not analyzed as time-dependent variable [15]. Unlikely, in dynamic modeling approach, the arc is simulated or formulated as variable time-dependent impedance [16]. For example, a dynamic model aiming prediction of DC flashover voltage of insulators considering variation of pollution layer resistance in wet period is proposed and a mathematical exponential formulation is as well represented in [14] to calculate pollution layer resistance by time. A multi-arcs dynamic approach is adopted in [9], [17] to represent a flashover prediction model by consideration of combination of several dry bands and polluted bands.

Recently, dynamic flashover modeling by use of intelligent algorithms (IAs) e.g., genetic algorithm (GA) in [6], artificial neural network (ANN) in [18]–[20], and fuzzy logic (FL) in [21] has been proved useful to study pollution behavior of insulators with better correlation to real experimental data of polluted insulators. Some of previously presented works have tried to predict critical pre-flashover conditions explicitly in order to avoid failure of insulators in harsh conditions without considering analytical formulations on levels of development of dry band arcing on the surface of polluted insulators leading to flashover such as [18], [22], [23]. Whereas, some other researches have focused on understanding levels of propagation of arcs in order to make correlation in terms of mathematical circuit formulations aiming to predict flashover voltage (FOV) such as [6], [9], [24].

Main purpose of this paper is to represent a dynamic modeling of voltage of polluted insulator as voltage of pollution region and voltage of discharge region. It is shown that investigation of these voltages can be used to predict voltage and time of flashover in polluted insulators. Due to stochastic nature of discharges on surface of polluted insulators, Gaussian modeling is used to represent analytical formulations of pollution region and discharge region. Finally, critical pre-flashover conditions of real testing polluted insulators are investigated according to

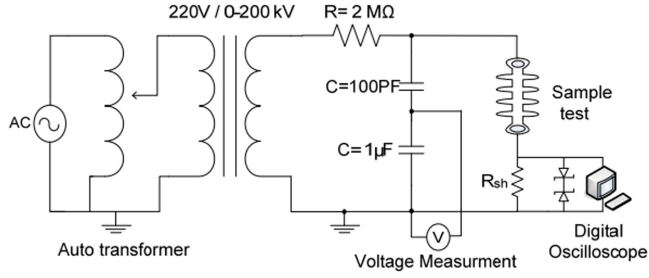


Fig. 1. Schematic view of test-setup, measurement circuit of voltage and leakage current [25].

calculated critical points of Gaussian representations of pollution voltage and discharge voltage and prediction of voltage and time of flashover are obtained. This procedure of flashover prediction can be used to develop an early warning system in online monitoring of in-service insulators in order to prevent pollution flashover.

II. EXPERIMENT SETUP

Laboratory of Polymer Insulators is part of High Voltage Institute of the University of Tehran; which the designed pollution flashover tests are performed there. The general schematic diagram of the experimental setup is shown in Fig. 1 [25]. Setup of the pollution flashover test of polymer insulators consists of a chamber that is made from galvanized metal sheets with dimensions of 2m*2m*2m. The testing chamber is electrically earth-wired for safety measures. In addition, a step-up 100 kV transformer, and a humidifier which consists of an electric water pump and standard nozzles are, other main component of the test setup. The applied voltage was measured by a capacitive voltage divider, with a voltage ratio of 10000:1 and the LC was measured by means of the voltage of the shunt resistor in series with the insulators [25]. All the samples are selected from over-head 20 kV distribution feeders of south of Iran (polymer insulators with the same silicone rubber material). Age of samples is 5 years. Hydrophobicity classes of the samples based on IEC62073 [26] is HC3. It means that the insulators are neither completely hydrophobic nor hydrophilic. Hence, hydrophobicity of the samples is moderate.

In each test, the applied voltage is gradually increased. At the time of $t = 0$, the input peak voltage is set to 10 kV (7.14 kV RMS value); then the peak voltage would be increased by step of 1 kV (or 0.71 kV RMS) per minute, while the voltage and LC waveforms were recorded. During the test, pollution is sprayed into the test chamber with the constant equivalent salt deposit density (ESDD) of a solution comprised by distilled water and NaCl. The electrical conduction of the pollution solution is measured and kept constant as a non-variable parameter in each test. The data acquisition system consists of a PC, a digital oscilloscope and a shunt resistors box and a voltage divider. The reported file for all measured variables can be analyzed through favorable algorithms and formulations in MATLAB software.

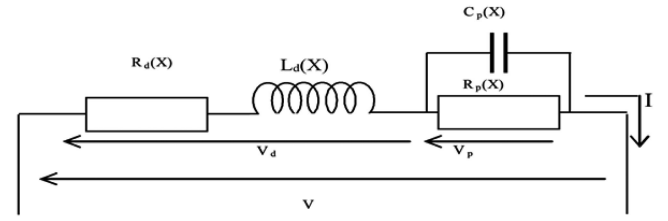


Fig. 2. Equivalent circuit model [27].

III. DYNAMIC MODEL OF BREAKDOWN

Simply in order to elaborate dynamic model of the arc, a cylindrical-cross section model of arc is investigated which consist of pollution layer part and discharge layer (discharge band) part which is called Obenaus's model [8] and has analyzed polluted surface for the first time, and also is suitable for precise analysis of conditions of permanent discharge formation on the surface of insulators. This model is shown in Fig. 2 [27]. As it can be seen from the figure, total voltage on the insulator can be divided to three parts as it follows:

$$U_{ins} = U_E + U_d + U_p \quad (1)$$

which U_{ins} , U_E , U_d , U_p are total applied voltage to insulator, voltage losses on electrodes, voltage of the electrical discharge part, and voltage of the wet polluted part, respectively. U_E is negligible, and can be ignored. Hence, the circuit voltage can be considered as sum of U_d and U_p [12], [27]. Proposed model in this research is included 4 parts. First part is the investigation of analytical behavior of Obenaus's model, from which the important parameters is extracted, and through analysis of these parameters, U_d , U_p and U_{ins} can be calculated. In second part, values of the parameters of the first part are obtained using experimental data on testing insulators and by means of a proposed intelligent algorithm. In third part, output results of the intelligent algorithm are analyzed, and a dynamic model is proposed by use of analysis of voltage behaviors of wet polluted layer (U_p) and discharge layer (U_d). Finally in last part, this dynamic model is validated on several real samples with various conditions.

The circuit model that is shown in Fig. 2 can be expressed as following equation based on [6], [8]:

$$Z_{eq} = \frac{U_{ins}}{I_d} \quad (2)$$

which Z_{eq} is the equivalent impedance of the whole insulator. In [27], it is mentioned that the existed inductor in equivalent circuit due to discharge of the insulator has a small value which can be neglected in calculations. Hence, its effect to final results can be ignored and it does not influence the following analysis. The above equation can be written as [8]:

$$Z_{eq} = R_d + \frac{1}{\frac{1}{R_p} + j\omega C_p} \quad (3)$$

or:

$$Z_{eq} = \left(R_d + \frac{R_p}{1 + (\omega R_p C_p)^2} \right) + j \left(-\frac{R_p^2 C_p \omega}{1 + (\omega R_p C_p)^2} \right) \quad (4)$$

As it can be seen from the above equation, Z_{eq} has two terms including real term and imaginary terms as Z_{real} and Z_{img} , respectively [8]:

$$Z_{eq} = Z_{real} + jZ_{img} \quad (5)$$

which [8]:

$$Z_{real} = R_d + \frac{R_p}{1 + (\omega R_p C_p)^2} \quad (6)$$

$$Z_{img} = -\frac{R_p^2 C_p \omega}{1 + (\omega R_p C_p)^2} \quad (7)$$

Real term and imaginary term of the equivalent impedance can be found using three parameters: applied voltage to the insulator, leakage current (LC) and phase shift between these two parameters. Hence, the (6) and (7) can be written as [8]:

$$Z_{real} = \left(\frac{U_{ins}}{I_d} \right) \cos(\theta) \quad (8)$$

$$Z_{img} = \left(\frac{U_{ins}}{I_d} \right) \sin(\theta) \quad (9)$$

which θ is the phase shift between applied voltage to the insulator (U_{ins}) and LC (I_d). The following equation can be written using (6) and (7) [8]:

$$R_p C_p \omega = \frac{Z_{img}}{R_d - Z_{real}} \quad (10)$$

hence:

$$C_p = \left(\frac{1}{R_p \omega} \right) \left(\frac{Z_{img}}{R_d - Z_{real}} \right) \quad (11)$$

Then, using (6) and (11), following equation can be written [8]:

$$Z_{real} - R_d = \frac{R_p}{1 + \frac{Z_{img}^2}{(R_d - Z_{real})^2}} \quad (12)$$

hence:

$$R_p = \frac{(R_d - Z_{real})^2 + Z_{img}^2}{Z_{real} - R_d} \quad (13)$$

Above equations show the relationships between circuit models of R_p , C_p and R_d which an approach to calculate these parameters should be proposed at first step using intelligent algorithm.

A. Hybrid Genetic Algorithm and Particle Swarm Optimization (HGAPSO)

The hybrid GA-PSO (hybrid genetic algorithm-particle swarm optimization), algorithm, which is used in this research, is developed by make use of the advantages of GA and particle swarming optimization (PSO) simultaneously [28]. The hybrid approach proceeds with half the total number of iteration initially

being performed by GA, and the optimum solution among these iteration then being fed into the PSO as the initial population. Thus, the near-optimal solution assigned by GA prevents PSOs from wasting resources on poor individuals. It can achieve a faster convergence rate than the standard PSO algorithm. The performance of the algorithm is described as follows [29]:

Step 1: Initialize each variable of GA.

Step 2: Initialize each variables of PSO.

Step 3: Arrange the group of PSO and calculate the coefficient of each condition of optimal solution of GA.

Step 4: Introduce optimal position condition to GA and perform crossover and mutation in GA.

Step 5: Take the best solution as the final solution and stop if the best individual in GA or PSO meets the termination criterion.

Step 6: If condition of PSO is satisfied with target condition (iteration number or target value), reproduction procedure is halted. Otherwise, it goes to step 3.

Step 7: Perform hybrid process if generation could be divided exactly by the designated iterative times N. Select P individuals from both sub-systems randomly according to their fitness and exchange. Go to step 2.

Based on intelligent algorithm (IA), dynamic modeling of pollution arc is developed in two parts. Part one is measurement of values of circuit parameters, and calculation of objective functions which are values of U_p and U_d by HGA-PSO. In part two, by means of proposed dynamic model, prediction of complete discharge and FOV is done by means of analytical behaviors of U_p , U_d and other parameters in different pollution conditions. In following sections, general structure of HGA-PSO is explained, and performance of this algorithm on defined problem of this research is investigated and discussed as well.

B. Calculation Circuit Parameters

In order to calculate values of U_p and U_d in testing insulators, values of circuit parameters should be first calculated and then, values of U_p and U_d can be calculated by use of calculated values of R_d , R_p and C_p . As it was explained in previous section, HGA-PSO is an optimization algorithm that can solve equations with several unknown variables provided that proper objective functions are defined for them. Values of Z_{real} and Z_{img} related to the circuit that is shown in Fig. 2 can be obtained using (8) and (9), since values of U_{ins} , LC and θ are known by use of data of experiments. Hence, objective function can be defined using (7) as follows:

$$Z_{img} + \frac{R_p^2 C_p \omega}{1 + (\omega R_p C_p)^2} = 0$$

$$\rightarrow \text{cost}(1) = \left| Z_{img} + \frac{R_p^2 C_p \omega}{1 + (\omega R_p C_p)^2} \right| \quad (14)$$

which $\text{cost}(1)$ in above equation is a cost function as objective function of optimization algorithm and unknown values of R_p and C_p can be calculated by minimizing the objective function using the optimization algorithm. Another objective function

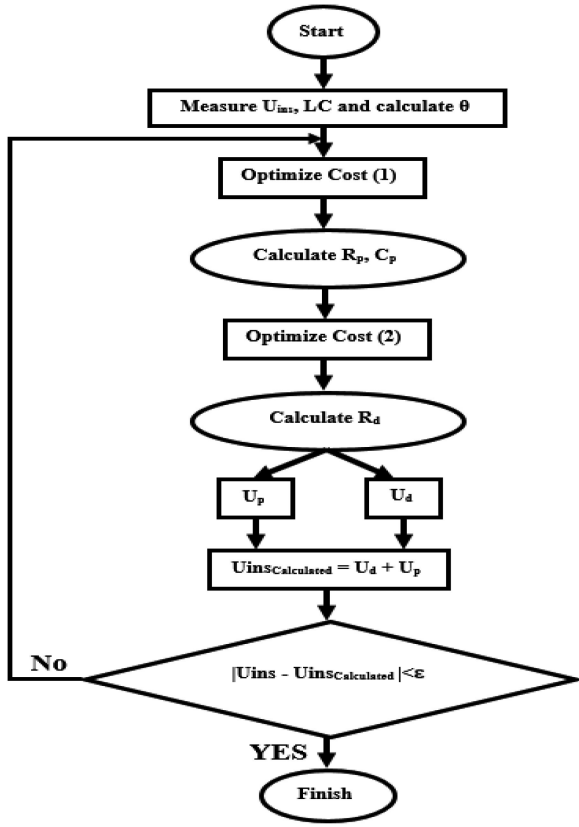


Fig. 3. Flowchart of calculating values of U_p and U_d using intelligent algorithm.

can be also defined using (13) as follows in order to calculate instantaneous values of R_d :

$$R_p - \frac{(R_d - Z_{real})^2 + Z_{img}^2}{Z_{real} - R_d} = 0$$

$$\rightarrow \text{cost (2)} = \left| R_p - \frac{(R_d - Z_{real})^2 + Z_{img}^2}{Z_{real} - R_d} \right| \quad (15)$$

After calculation of values of R_p , C_p and R_d , values of U_p and U_d can be calculated as following expressions, and then can be validated by use of U_{ins} :

$$U_p = (R_p || C_p) * I_d = \left(\frac{R_p * \frac{1}{j\omega C_p}}{R_p + \frac{1}{j\omega C_p}} \right) * I_d$$

$$\rightarrow U_p = \frac{R_p}{1 + j\omega R_p C_p} * I_d$$

$$\rightarrow U_p = \frac{R_p * (1 - j\omega R_p C_p)}{1 + (\omega R_p C_p)^2} * I_d \quad (16)$$

$$U_d = R_d I_d \quad (17)$$

Flowchart that is shown in Fig. 3, shows steps of measuring and calculating values of U_p and U_d using intelligent algorithm. Threshold value ε is $10^{-4} * U_{ins}$ in all calculations. This value

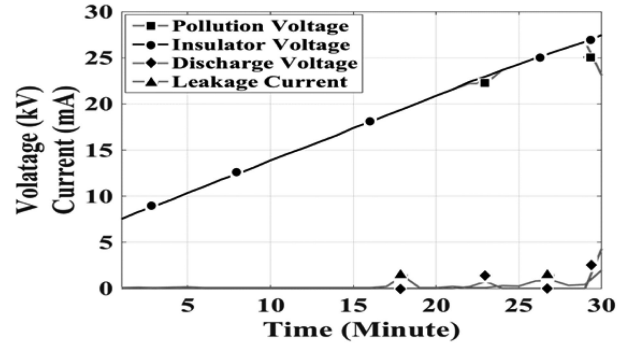


Fig. 4. RMS Values of U_p , U_d , U_{ins} and the LC of insulator for first zone.

is chosen through trial and error. It is tried to choose this value in order to optimize calculation speed according to hardware of computer system.

C. Analysis of Results

According to (8) and (9), it is observed that circuit models of Z_{real} and Z_{img} are functions of phase shift (θ), and variations of θ results in changes of values of Z_{real} and Z_{img} , which results in changes of values of R_p and R_d , and consequently changes of values of U_p and U_d . According to variations of U_p and U_d , and comparing them to each other, and their comparison to voltage of insulator (U_{ins}) as well, voltage behavior of insulator can be categorized into 4 zones, which by means of a suitable intelligent algorithm can predict permanent discharge (flashover) in polluted insulators. According to IEC 60815 [30], pollution on the surface of insulator can be classified into 5 regions as very light, light, medium, high and very high. This classification is based on the intensities of pollution deposits as equivalent soluble deposit density (ESDD) and non-soluble deposit density (NSDD). In this research, only used pollution is the solution of NaCl with distilled water as ESDD testing parameter. Values of ESDD in tests of this research are 0.01 mg/cm^2 , 0.1 mg/cm^2 , 0.2 mg/cm^2 , and 0.5 mg/cm^2 which are considered to be light, medium, high and very high, respectively based on IEC standard (In the very light region, value of LC is very low; hence it is ignored). In all tests, contamination process is done based on section (II) and surface conduction of the pollution solution is measured by means of conduction metering device which is converted into equivalent ESDD as mg/cm^2 by use of surface polluted area.

1) *First Zone: Light Level Pollution:* In first zone, ESDD of pollution solution is applied equal to 0.01 mg/cm^2 in test of this research, which is considered to be light level pollution. During the test, pollution level is kept constant while instantaneous voltage and LC of the insulator is measured and recorded. Then, values of U_p and U_d is calculated using proposed algorithm; RMS values of U_p , U_d , U_{ins} and the LC of insulator are shown in Fig. 4 (Peak value of U_{ins} is 10 kV and RMS value is 7.14 kV). As it can be seen from the figure, voltage trends of U_p and U_{ins} are similar and follow each other; the difference is seen in ending times of the test which U_d starts to increase as the U_p decreases nearly the same level. U_d is dependent on R_d and LC. In first region, value of LC is very small. That is because

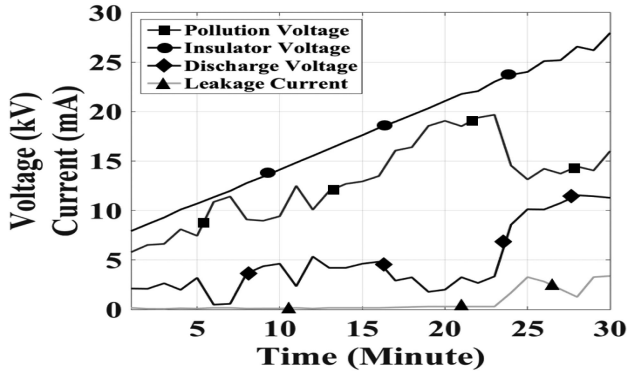


Fig. 5. RMS Values of U_p , U_d , U_{ins} and the LC of insulator for second zone.

of low level pollution which in combination of distilled water results in low surface conduction. In addition, dynamic behavior of R_d is somewhat complicated and it is usually a function of pollution level, wetting rate, arcing length, and LC. At final times of test in first zone of pollution level, low surface conduction would be existed due to high wet condition which results in more conduction level and LC formation in the insulator, and finally appearance of U_d . As a result, at the end of this time, some small discharges as dry band arcing can be observed through investigation of LC due to evaporation of surface water.

2) *Second Zone: Medium Level Pollution:* In second zone, ESDD of pollution solution is applied equal to 0.1 mg/cm^2 in test of this research, which is considered to be medium level pollution. After passing some time in first zone, due to higher level of pollution and more level of wet condition, the insulator enters the second zone, actually, in order to make change in surface pollution of insulators according to the test standard [31], with increase of pollution level and preparation of the standard, entering to the next zone would be done. In this zone, the variations of U_d are more remarkable, and the values of U_d is increased remarkably. This is shown in Fig. 5. As it can be seen from this figure, at some point of the curve, a sudden increase in value of U_d occurs. This is accompanied by a partial increase in LC (almost 3 mA). But, it can be observed that U_d cannot be greater than U_p in this wet polluted period of test. In fact, in early times of the test, U_d is strongly a function of R_d . Due to pollution solution on the surface, surface conduction is tending to increase because of wetting condition. This results in development of R_d , even though value of LC is low. After about 17th minute, development of LC with low speed can be seen from Fig. 5, which results in evaporation of surface water and dry band formation. Consequently, R_d decreases and U_d decreases remarkably as well (the value of LC is not strong enough to increase the U_d). However, after 23th minute, development of dry band arcing results in high value of LC which supports development of U_d . From that time, it is LC that causes variation of U_d .

In early moments of the test, surface pollution with combination of wetting condition results in a conductive solution on the surface in which insulator would shift from an absolute capacitive form to a resistive-capacitive form. This process can

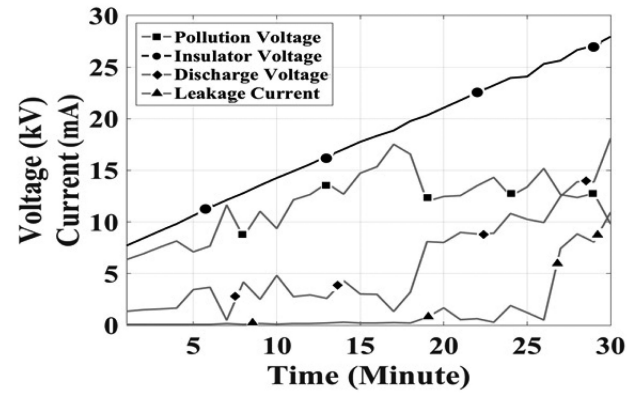


Fig. 6. RMS Values of U_p , U_d , U_{ins} and the LC of insulator for third zone.

be followed by increase of θ . That is why R_d is observable in the real part of Z_{eq} at this time. Formation of R_d would results in appearance of U_d and LC. As wetting condition and consequently surface conduction increases, LC grows in a way that is able to vaporize some of the water on the surface. Hence, some dry spots would be formed on the surface of insulator which causes further development of U_d and LC. However, in second zone, these dry band arcing are not much enough to develop U_d in a way that critical condition of $U_d > U_p$ is achieved. However, sudden change of U_d is occurred in this zone as discussed above can be used as warning signals to predict discharge in insulator.

3) *Third Zone: High Level Pollution:* In third zone, ESDD of pollution solution equal is applied to 0.2 mg/cm^2 in test of this research, which is considered to be high level pollution. The distinguishing border between zone 2 and zone 3 can be seen from Fig. 6, in which U_d is starting to get higher values than U_p . It can be observed from Fig. 6 that in 26th minute of the test, the value of U_d becomes more than that of U_p , and this results in increase of LC to 7 mA in later close moments. In fact, two important events occurred simultaneously; first was sudden increase of U_d and $U_d > U_p$, and second was remarkable increase of LC in wet polluted insulator, in which these two events can be used as warning signals. Consistent formations of dry band arcing due to frequently increase and decrease of temperature LC is as well another observation in this period of testing.

4) *Fourth Zone: Very High Level Pollution:* In fourth zone, ESDD of pollution solution is applied equal to 0.5 mg/cm^2 , which is considered to be very high level pollution. As it will be shown and discussed, behavior of the wet polluted insulator results in complete flashover in this zone. Similar to other discussed zones, curves of U_p , U_d and U_{ins} , and curve of LC of insulator are shown in Fig. 7. As it can be seen from this figure, dynamic behavior of the insulator is primarily like zone 3, the difference is that the amount of discharge voltage is higher than the pollution voltage.

Investigating several tests with very heavy level of pollution show that number of dry band arcing in this pollution level cannot be compared to other levels (it is too much). As a result, it can be said that powerful dry band arcing with combination of very

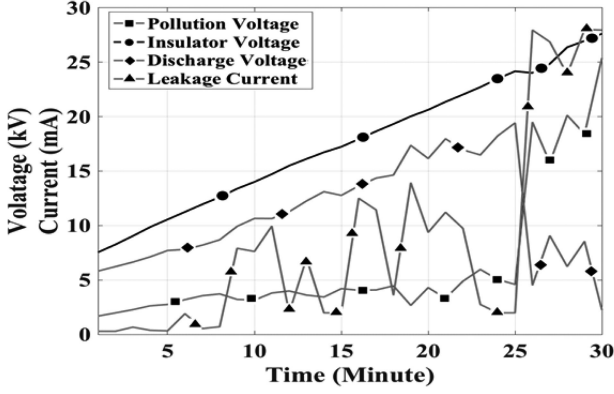


Fig. 7. RMS Values of U_p , U_d , U_{ins} and the LC of insulator for fourth zone.

high level of surface conduction results in rapid increase of LC. This is the beginning for complete discharge on the surface of the insulator. As it can be seen from the figure, after 15th minute higher values of LC (more than 10 mA) exist which can be the beginning of complete discharge on surface of insulators. Summarily, LC due to dry band arcing in zone 4 can have more than twice the similar value in zone 3 by average (In this zone, it should be noted that the condition $U_d > U_p$ is always valid. In addition, the average value of LC for dry band arcing model in zone 4 is 10 mA and in zone 3 is 5 mA). As a whole, it can be claimed that represented model is able to predict dry band discharge before flashover, and it can be used as a signal warning system in real-time monitoring. However, in 25th minute of the test, two remarkable events occur; first is sudden change of values of U_p and U_d as $U_p > U_d$, which is consistent, and second is that LC tends to increase to high values. These two events occur at the time that insulator is on the verge of complete flashover, as the value of LC even passes over 25 mA.

IV. MODEL OF INSULATOR BASED ON GAUSSIAN MODEL

Due to random nature of measured data concerning tests of polluted insulators, in this research, an analytical model based on probability functions is defined in order to model 4 zones which were discussed in previous sections. One of useful function that is used in probabilistic mathematics is Gaussian function. Whenever studied data show a behavior like normal function, modeling functions as Gaussian function can present suitable results [32]. In addition, according to noise-like behavior in data of the tests of studied insulators which show random behavior, Gaussian model can be considered as a suitable candidate to represent behavior of insulators. Gaussian model is applied to model of U_p and U_d for four zones which were explained in earlier sections. Due to variable nature of U_p and U_d and their dependence to test parameters such as wetting rate, electrical conduction value and testing time, second-order Gaussian function is used to obtain a more accurate model of U_p and U_d . Second-order Gaussian function is expressed as follows: [33]

$$f(t) = k_1 e^{\left(-\frac{(t-\mu_1)}{\sigma_1}\right)^2} + k_2 e^{\left(-\frac{(t-\mu_2)}{\sigma_2}\right)^2} \quad (18)$$

TABLE I
VALUE OF GAUSSIAN FACTORS OF ZONE 1

Variable	U_p	U_d
k_1	$1.61 \cdot 10^4$	$9.73 \cdot 10^{17}$
k_2	$1.87 \cdot 10^4$	$1.11 \cdot 10^3$
μ_1	31.18	30.96
μ_2	17.15	23.71
σ_1	8.85	0.87
σ_2	16.89	0.53

TABLE II
VALUE OF GAUSSIAN FACTORS OF ZONE 2

Variable	U_p	U_d
k_1	$4.7 \cdot 10^3$	$1.43 \cdot 10^8$
k_2	$1.71 \cdot 10^4$	$9.77 \cdot 10^3$
μ_1	24.56	227.41
μ_2	26.99	3.72
σ_1	3.43	64.51
σ_2	27.61	1.97

TABLE III
VALUE OF GAUSSIAN FACTORS OF ZONE 3

Variable	U_p	U_d
k_1	$6.22 \cdot 10^3$	$6.99 \cdot 10^3$
k_2	$1.41 \cdot 10^4$	$8.44 \cdot 10^3$
μ_1	20.03	27.41
μ_2	26.42	60.28
σ_1	2.94	3.92
σ_2	29.72	47.24

TABLE IV
VALUE OF GAUSSIAN FACTORS OF ZONE 4

Variable	U_p	U_d
k_1	$2.17 \cdot 10^4$	$1.52 \cdot 10^4$
k_2	$6.34 \cdot 10^3$	$1.59 \cdot 10^4$
μ_1	39.61	30.95
μ_2	20.74	18.27
σ_1	4.71	7.34

According to expression (18), Gaussian model of four zones can be formulated as Tables I to IV. As it was observed in Fig. 4 to 7, U_p and U_d behavior are investigated and using the mentioned Gaussian modeling, the appropriate curve fitting is used to obtain the appropriate coefficients. Calculated values of Gaussian factors as k_1 , k_2 , μ_1 , μ_2 , σ_1 and σ_2 are shown in Tables I to IV.

V. DISCUSSION

A. Proposed Flashover Model

As it was discussed before, Figs. 4 to 7 show dynamic behavior of U_d and U_p which are obtained by curve fitting of circle curves (U_d) and diamond curves (U_p) based on Gaussian modeling according to (18) and Tables I to IV, are shown in Fig. 8. As it can be seen from this figure that circle-curves and diamond-curves are obtained for U_d and U_p , respectively. In other words, instead of Figs. 4 to 7 which show real dynamic behavior of U_d and U_p , Gaussian models of U_d and U_p are drawn in Fig. 8 which can be used to predict voltage and time of flashover.

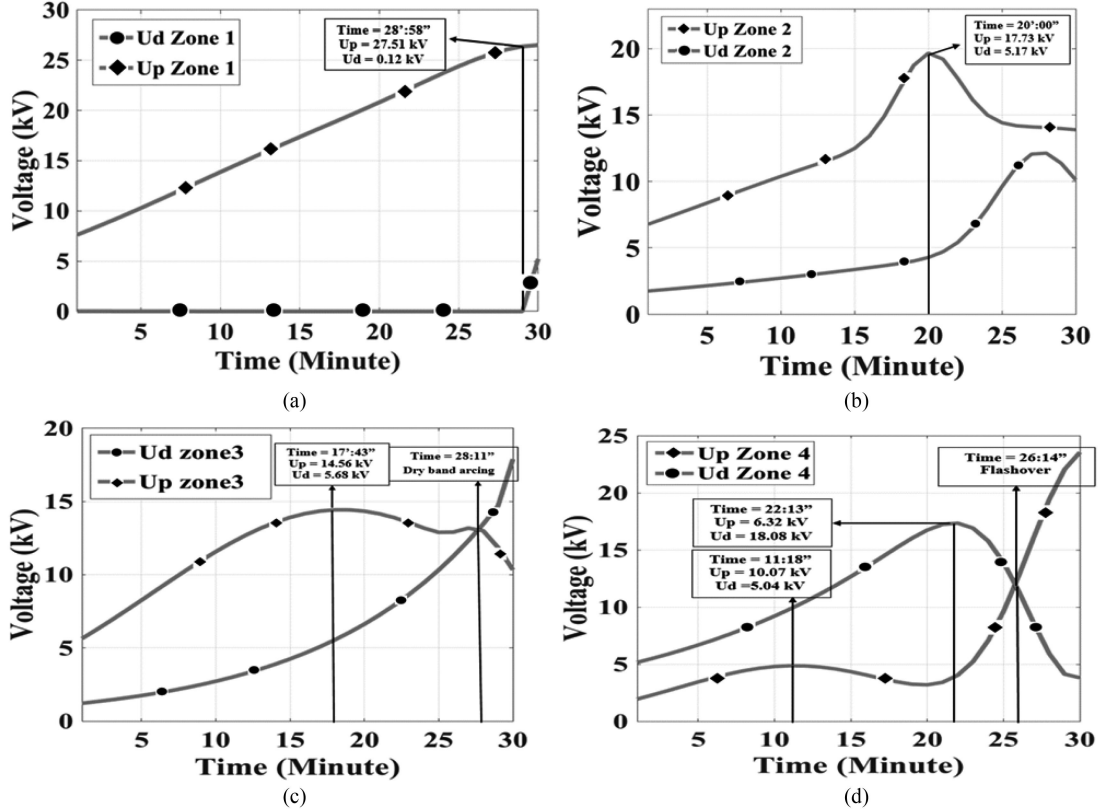


Fig. 8. Variation trend of U_p and U_d based on Gaussian model, in (a) Light level pollution zone. (b) Medium level pollution zone. (c) High level pollution zone. (d) Very high level pollution zone.

Proposed Gaussian model is shown on Fig. 8(a) at beginning time of the test. It is observed that with light level pollution, as the time of test passes, variation trend of U_p and U_d is not significant; however an extreme point can be observed from the curve of U_p at the ending time of the test which this maximum point simultaneously results in start of dry band arcing in tested insulator. This is recorded at the time $t = 28'58''$ and at the voltage $U_p = 27.51 \text{ kV}$ which causes low LC flow on the surface of insulator that is shown in Fig. 4. As it was discussed, at ending times of low level pollution test, small activities in behavior of U_p and U_d and LC as well were occurred. In second zone that is in medium level pollution zone, it is observed that activities of discharges and variation trend of U_p and U_d are started sooner in middle times of the test. Variation trend of U_p and U_d , in medium level pollution zone, are shown in Fig. 8(b). As it can be seen from the figure, maximum point of U_p is occurred at the time $t = 20'00''$ while this is the time that U_d starts to increase.

Investigating Fig. 8(c) (based on Table III), in third pollution zone, reveals similar results to second zone, in which maximum point of U_p can lead to prediction of dry band arcing in tested insulators before the time of $U_d > U_p$. These conditions in all three zones occur before U_d consistently gets higher values than U_p . Based on Gaussian model and calculated functions, at the time $t = 17'43''$, value of U_p reaches to its maximum point which is $U_p = 14.56 \text{ kV}$, while at this time $U_d = 5.68 \text{ kV}$. This moment shows peak point of U_p ; and in fact, it is the starting point of dry band arcing which can be observed according to Fig. 8(c)

at time of $U_d > U_p$, $t = 28'11''$. Variation trends of U_p and U_d in fourth pollution zone, which present dynamic behavior of insulator in flashover condition, show some notifications unlike to low level pollution region; as U_d in very high level pollution zone dynamically behaves like U_p in low level pollution zone.

Therefore, maximum point of U_d and the flag signal $U_p > U_d$ should be searched in this zone to predict flashover of insulator. U_p and U_d versus time of test based on Gaussian fitness according to Table IV are shown in Fig. 8(d). It is observed that in $t = 22'13''$, U_d reaches to its maximum point as $U_d = 18.08 \text{ kV}$. At this point, $U_p = 6.32 \text{ kV}$ and U_p is ready to increase. Hence, this point can be regarded as critical point to predict flashover in tested insulator. After this point, decreasing U_d while increasing U_p leads to attainment of $U_p > U_d$ and consequently flashover of tested insulator that is accompanied by sudden increase in LC. This is occurred at $t = 26'14''$ and in fact, it is beginning of flashover.

B. Model Validation

Summarily, according to investigation of proposed model in order to predict pollution flashover in tested insulator, it is observed that two critical points in dynamic behavior of U_p and U_d are significant: first calculated point results in prediction of dry band arcing that is shown in Fig. 9(a), and second point results in prediction of flashover that is shown in Fig. 9(b).

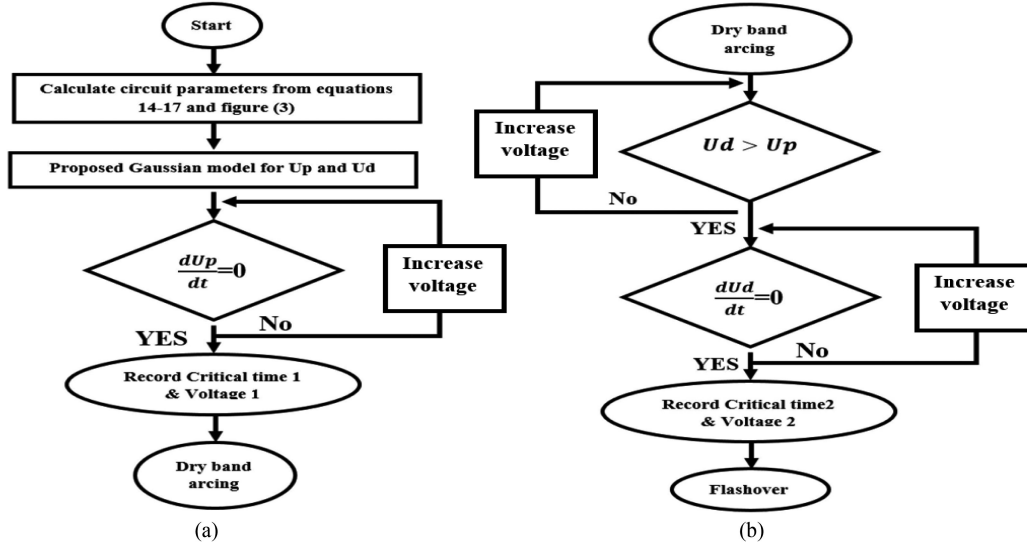


Fig. 9. Flowchart of calculating critical time and voltage of dry band arcing (a) and flashover (b).

TABLE V
RESULTS OF THE PROPOSED GAUSSIAN MODEL

ESDD (mg/cm ²)	Critical Time 1	U_{ins1}	Critical Time 2	U_{ins2}
0.01	28':58"	27.63 kV	-	-
0.1	20':00"	22.49 kV	-	-
0.2	17':43"	20.24 kV	-	-
0.5	11':18"	15.11 kV	22':13"	24.40 kV

TABLE VI
RESULTS OF THE REAL TEST

ESDD (mg/cm ²)	Critical Time 1	U_{ins1}	Critical Time 2	U_{ins2}
0.01	29':00"	30 kV	-	-
0.1	23':00"	23 kV	-	-
0.2	26':00"	22 kV	-	-
0.5	15':00"	17 kV	25':00"	25 kV

As it was discussed in this section, it can be concluded that proposed model can achieve to predict two critical dynamic levels as dry band arcing and flashover through investigating variation trends of U_p and U_d . In the following, results based on proposed modeling, and results of the real test are respectively shown in Tables V and VI, in order to validate previous results. As it can be seen from Table V, instantaneous voltage of insulator is equal to sum of instantaneous U_p and U_d and is shown in critical time 1 and critical time 2 as U_{ins1} and U_{ins2} , respectively. All results are obtained based on proposed modeling. It is observed from the table that flashover time is only existed for very high pollution level. Also, increase of electrical conduction is led to decrease of time to reach critical points and decrease of required voltage to initiate dry band arcing. Next, as it can be seen from Table VI, four above-mentioned parameters are measured for the real test data of the polluted insulator: time of beginning of dry band arcing and its voltage, time of flashover and FOV. In these times, the applied voltage to the

TABLE VII
THE GEOMETRICAL PARAMETERS OF THE SAMPLES OF VALIDATION TESTS

Samples #	Arcing distance (cm)	Diameter of sheds (cm)	Num. of sheds	Creepage distance (cm)
A	30	14	5	65
B	35	10	7	70
C	30	14	4	58
D	28	10	5	53

tested insulator is recorded as well and peak value is increased by steps of 1 kV per minute (start peak voltage is 10 kV or RMS is 7.14 kV) until the conditions of dry band arcing and then flashover are attained, respectively. These recorded values are shown in Table VI. Comparing Tables V and VI shows a close correlation between results of simulation and results of real test data. This correlation shows that the proposed model is able to predict two significant dynamic behaviors of polluted insulators as time-variable before their occurrence. In fact, it is approved that relative extremes of U_p and U_d are effective tools for predicting times of dry band arcing and flashover.

In order to present a stronger validation, four types of silicone rubber insulator are selected and modeled with conditions of Table VII. Pollution test similar to very high level pollution are performed on these types of insulators in order to predict FOV. Also, real times of occurrence of beginning of dry band arcing and flashover are recorded (Based on Table VI). Then, average values and standard deviation of recorded times and voltages are made by following formulations:

$$Critical\ Time_1 = \frac{\sum_{i=1}^{10} Critical\ Time_{1i}}{10} \quad (19)$$

$$U_{ins1} = \frac{\sum_{i=1}^{10} U_{ins1i}}{10} \quad (20)$$

$$Critical\ Time_2 = \frac{\sum_{i=1}^{10} Critical\ Time_{2i}}{10} \quad (21)$$

TABLE VIII
THE RESULTS OF THE VALIDATION TESTS BASED ON PROPOSED MODEL

Samples #	Critical Time 1	U_{ins1}	Critical Time 2	U_{ins2}
A	21':54"	29.12 kV	25':31"	34.32 kV
B	23':17"	32.44 kV	26':12"	38.21 kV
C	19':22"	19.71 kV	22':14"	22.42 kV
D	16':12"	15.31 kV	21':28"	17.41 kV

TABLE IX
THE RESULTS OF THE VALIDATION TESTS BASED ON REAL TEST

Samples #	Critical Time 1	U_{ins1}	Critical Time 2	U_{ins2}
A	23':00"	30 kV	28':00"	35 kV
B	25':00"	34 kV	29':00"	40 kV
C	22':00"	21 kV	25':00"	23 kV
D	22':00"	18 kV	24':00"	21 kV

$$U_{ins2} = \frac{\sum_{i=1}^{10} U_{ins2i}}{10} \quad (22)$$

$$\sigma = \sqrt{\frac{\sum_{i=1}^{10} (x_i - \mu)^2}{10}} \quad (23)$$

Which σ , x_i and μ are standard deviation, each value from the population and the population mean respectively. Results of simulations based on proposed model, and results of real test data for four types of insulators are presented in Table VIII and IX, respectively. It can be seen from these tables that there are close correlations between simulation based on proposed model and real test data. According to the results of these tables, it is observed that in order to predict FOV based on proposed model, samples of A, B, C and D have predicted FOV of 34.32 kV, 38.21 kV, 22.42 kV and 17.41 kV, respectively; which have close correlations with measured voltages of the experiment (values are 35 kV, 40 kV, 23 kV and 21 kV, respectively). Also, calculated time to predict two critical times of beginning of partial discharges and complete discharge in very high level polluted insulators have close correlations with real times of their occurrences in experiment.

It can be seen from Table VII that tested samples have different profiles. An important point is that variation in features of profile of insulators result in variation of existing resistors and capacitors of insulators. Consequently, it is expected that this would change analytical circuit representation.

Figs. 10 and 11 show comparison of values of voltage and time of flashover between real test and proposed simulation. It can be seen from these figures that proposed procedure of flashover prediction has the least dependency on profile of insulators; this is because of the calculation process. Variation in profile of insulators would change existing resistors and capacitors of insulators but circuit elements are obtained based on instantaneous values of LC and voltage of insulators according to intelligent optimization algorithm that was expressed in section (III-B). Hence, dependency on profile of insulators is canceled and that is verified by results of Figs. 10 and 11. While investigating deviations of Figs. 10 and 11, it can be seen that how much difference

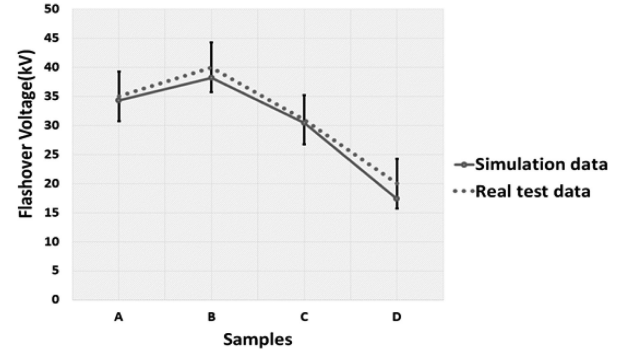


Fig. 10. Comparison of values of flashover voltage between real test and proposed simulation and standard deviation bars.

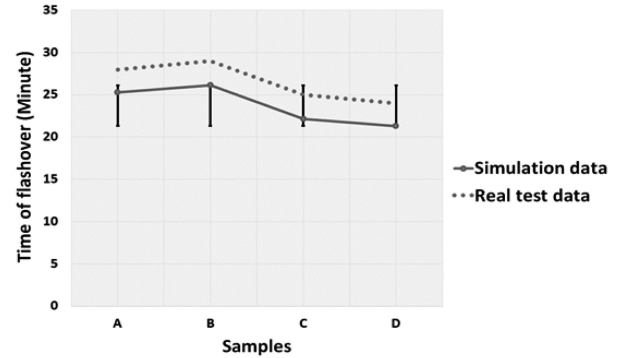


Fig. 11. Comparison of values of time of flashover between real test and proposed simulation and standard deviation bars.

TABLE X
THE RESULTS OF THE PREDICTION FOV BASED ON PROPOSED MODEL AND OTHER REFERENCES

Samples #	FOV based on [8]	FOV based on [16]	FOV based on proposed model	Real Test
A	32.88 kV	29.66 kV	34.32 kV	35 kV
B	37.01 kV	33.63 kV	38.21 kV	40 kV
C	22.58 kV	21.46 kV	22.42 kV	23 kV
D	16.10 kV	14.82 kV	17.41 kV	21 kV

exists between data and average values. This can be explained due to the oscillating and random nature of discharges; hence deviation in black curves of the two figures can be observed.

Prediction of FOV by use of the proposed model and previously presented models in [8] and [16] is shown in Table X. As it can be seen from the table, accuracy of the results by proposed model is better than other models. It should be noted that accurate prediction is achieved through solving the model by use of intelligent algorithm and independency of the calculations to profile of samples, pollution level, wetting rate and other testing parameters. In addition, Gaussian modeling played an important role to make better prediction of FOV and time of flashover in polluted insulators.

C. Real-Time Monitoring

It should be note, there are differences between real outdoor conditions and standard experimental conditions. According to

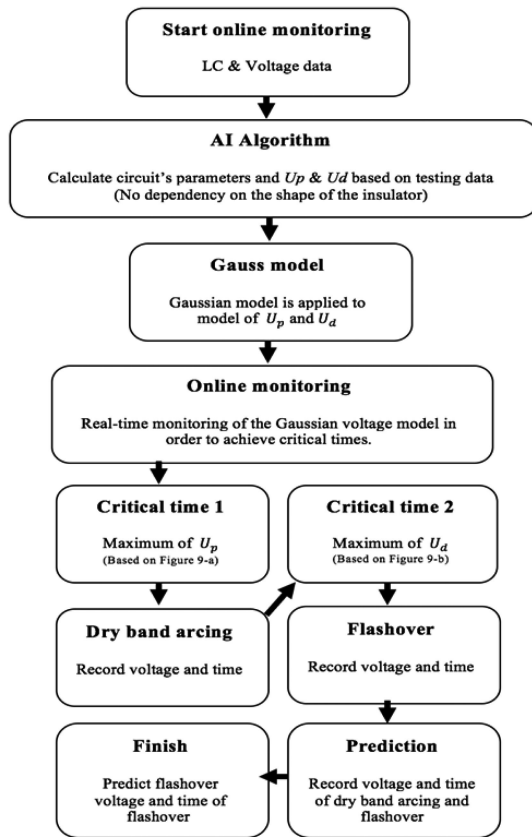


Fig. 12. Flowchart of real-time monitoring.

recent researches [6], [8]–[10], [12]–[16], synthetic accelerated tests are performed in experiment setup in order to investigate pre-flashover conditions of polluted insulators. It should be noted that all these researches are done based on standard protocols [26], [30], [31] about testing of insulators. For example, test types and synthetic pre-contamination methods in experimental environment are described in [31] in order to simulate real outdoor pollute conditions. Also, test conditions according to type of insulator, weather conditions, pollution type (such as salt-fog, industrial or iced) and other environmental conditions (such as pressure, temperature, wind, fog and etc.) are noticed in [30] standard so as to match experiment with real environment. Therefore, it is tried to make close test conditions to the real outdoor polluted conditions according to standard procedures. Finally, it should be noted that there are always some differences and errors between real outdoor conditions and synthetic pollution test conditions even though tests are performed based on standards. There are some noises on voltage and LC in real outdoor conditions. Even, there are some little noises in experimental environment (mostly on received voltage of insulators). This problem is solved through software post-process by use of proper filters in MATLAB codes. Hence, all noises on voltage and LC can be omitted through this way.

The proposed model can be used as an effective tool to predict both voltage and time of flashover of polluted insulators. In fact, in addition to the prediction of FOV, prediction of time of flashover is possible through proposed algorithm as well.

According to the obtained results, accuracy of the proposed model has the least dependency on profile of polymer insulator due to the solving process based on AI which can use real data of voltage and LC of any sample with different unknown profile. Therefore, this model can be developed for any polymer insulator based on proposed procedure. Hence, critical points of pre-flashover conditions can be calculated by online monitoring of LC and voltage and calculation of U_d and U_p so as to prevent pollution flashover of insulators. This process is summarized in Fig. 12 as a proposed flowchart of online monitoring. It can be seen from the figure that by real-time monitoring of leakage current and voltage, the equivalent circuit related to dynamic modeling approach can be obtained instantaneously through the proposed intelligent algorithm due to the changes of these two elements. Then, by calculating the circuit model and the behavior of the discharge voltage and the pollution voltage, Gaussian modeling can be developed and by analysis of the Gaussian model, aforementioned critical points related to the dry band arcing and finally flashover can be obtained. Finally, prediction of time of flashover and FOV is done.

VI. CONCLUSION

In this research, a dynamic approach was adopted to model pollution flashover conditions of polymer insulators. Experimental data of applied voltage and leakage current of polluted insulators were used to develop analytical circuit-based formulations. This was done by means of an intelligent PSO-HG algorithm. It was shown that through fitness of Gaussian model, dynamic behavior of flashover of polluted insulator from the beginning of partial discharges to the end of complete discharge can be explained. This is represented as a classification of different stages of pollution arcs in different conditions of pollution flashover tests. Two critical points of voltage and time are obtained based on proposed Gaussian model thorough analysis of discharge voltage and pollution voltage in order to predict dry band arcing and pre-flashover conditions. Close correlation is found between obtained simulated results and real test data. Also, prediction of FOV is compared between proposed model and previous models. Proposed model shows better correlation with real test data. Finally, an online monitoring system is proposed based on represented algorithm which can be used to develop an early warning system for prevention of flashover of in-service insulators.

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